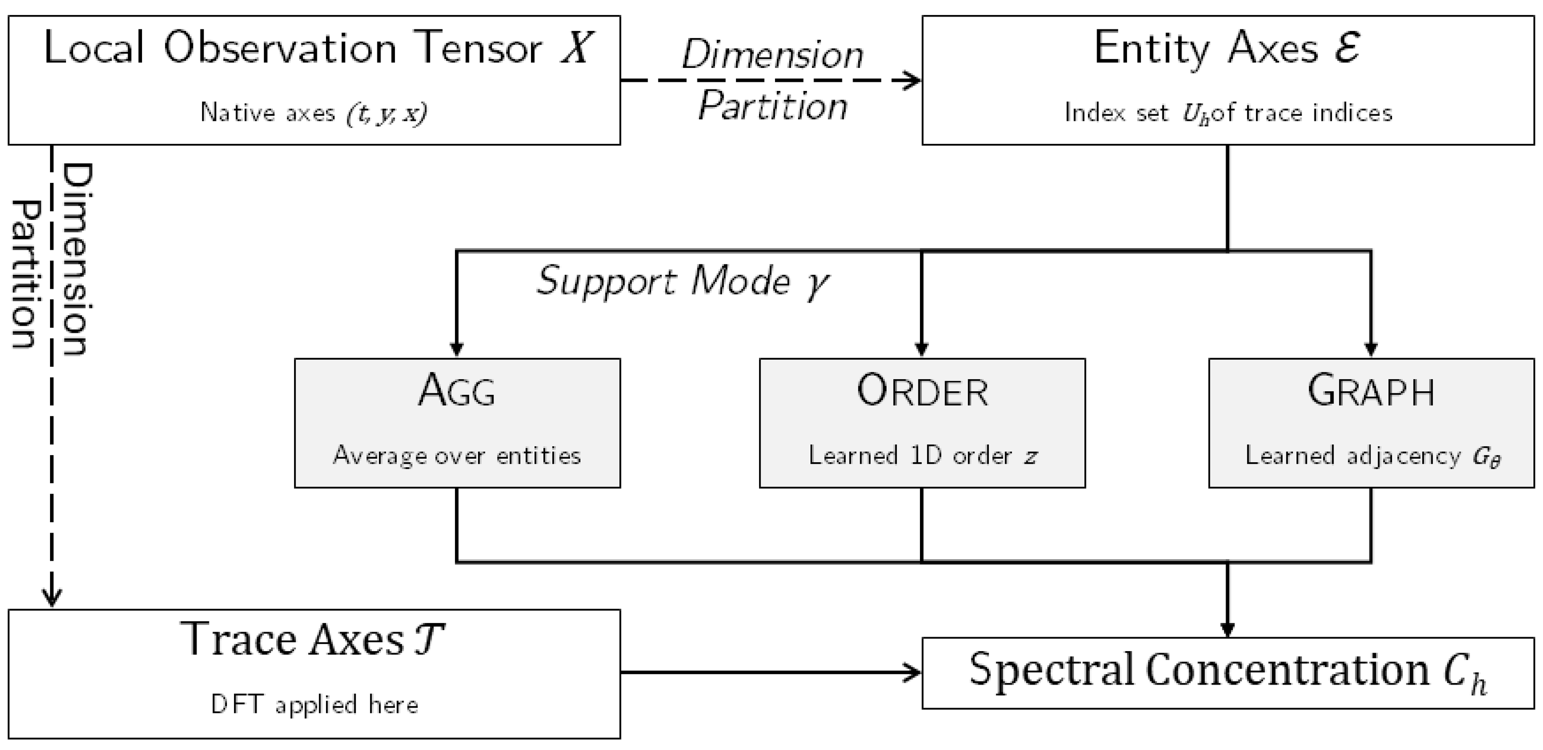
Observer-Based Pattern Discovery on Frozen Histories

Sebastian von Mammen, Tobias Buhl
University of Würzburg

Agnostic Observers Solve the Observational Problem

- **Observational Problem:** Deciding *whether* emergent patterns exist, *where* to look, and *which arrangement* forms a meaningful stage.
- **Risk of Fixed Supports:** Predefined viewpoints potentially obscure real patterns or manufacture spurious ones [1].
- **Agnostic Observers:** Adaptive local probes search simultaneously for patterns and reproducible viewpoints [2].
- **Downstream Benefits:** Localising mechanisms, finding explanatory patterns, and accelerating simulation via skip-ahead [3].
- **Frozen Histories:** Decoupling observer adaptation from live simulation avoids confounding patterns with single-run contingencies.

Dimension Partition & Support



Evolutionary Loop

- **Spectral Discovery Score**: Corrected spectral concentration $C_h^{(a)}$ on training anchors $\mathcal{A}_{\text{train}}$:

$$f_{\text{disc}}(h) = \frac{1}{|\mathcal{A}_{\text{train}}|} \sum_{a \in \mathcal{A}_{\text{train}}} C_h^{(a)} - \lambda_{\text{comp}} \mathcal{K}(h),$$

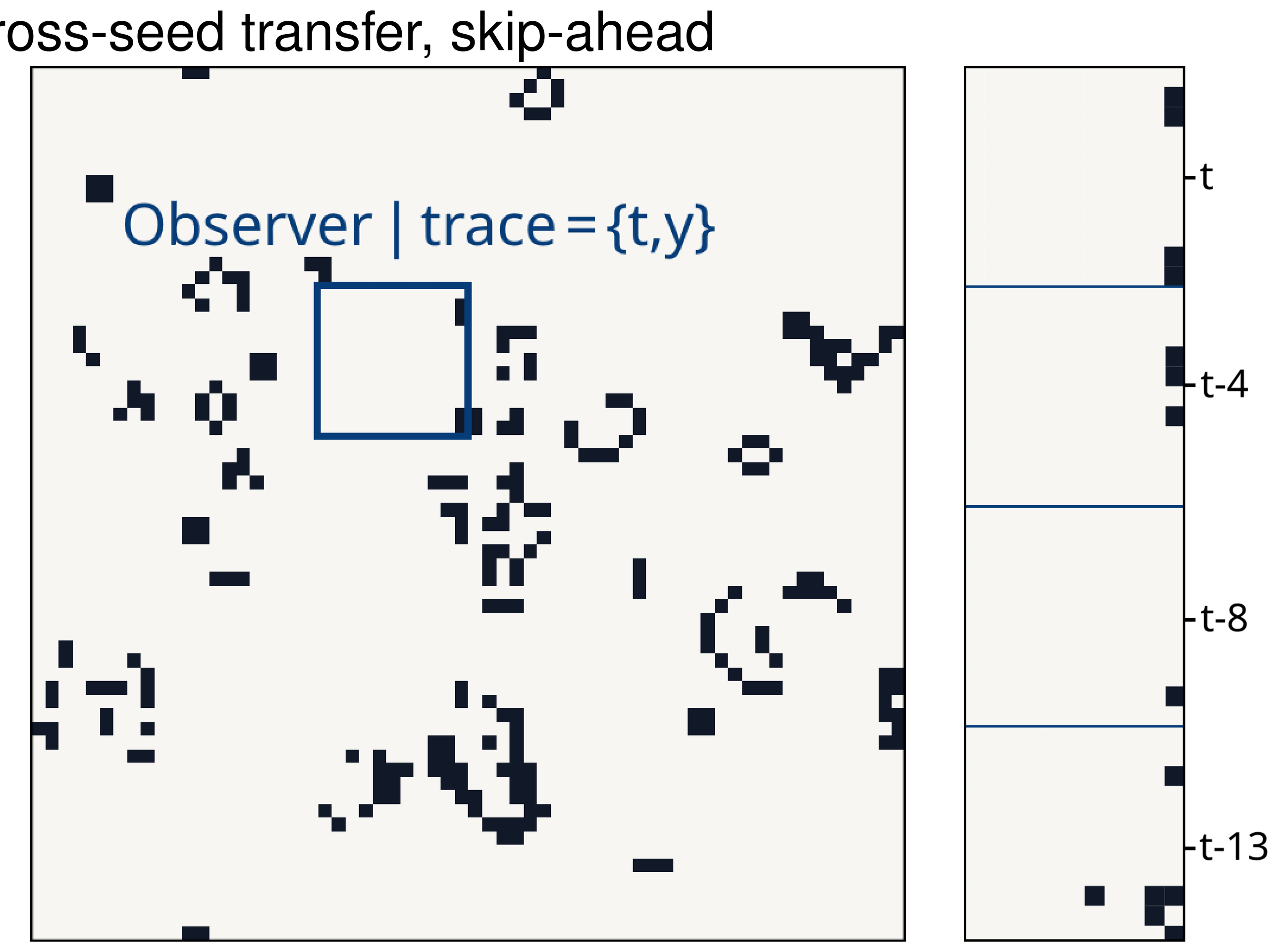
- **Predictive Score:** Chance-adjusted accuracy $A_{\text{adj}}^{(a)}$ on held-out validation anchors $\mathcal{A}_{\text{val}}$:

$$f_{\text{pred}}(h) = \frac{1}{|\mathcal{A}_{\text{val}}|} \sum_{a \in \mathcal{A}_{\text{val}}} A_{\text{adj}}^{(a)}$$

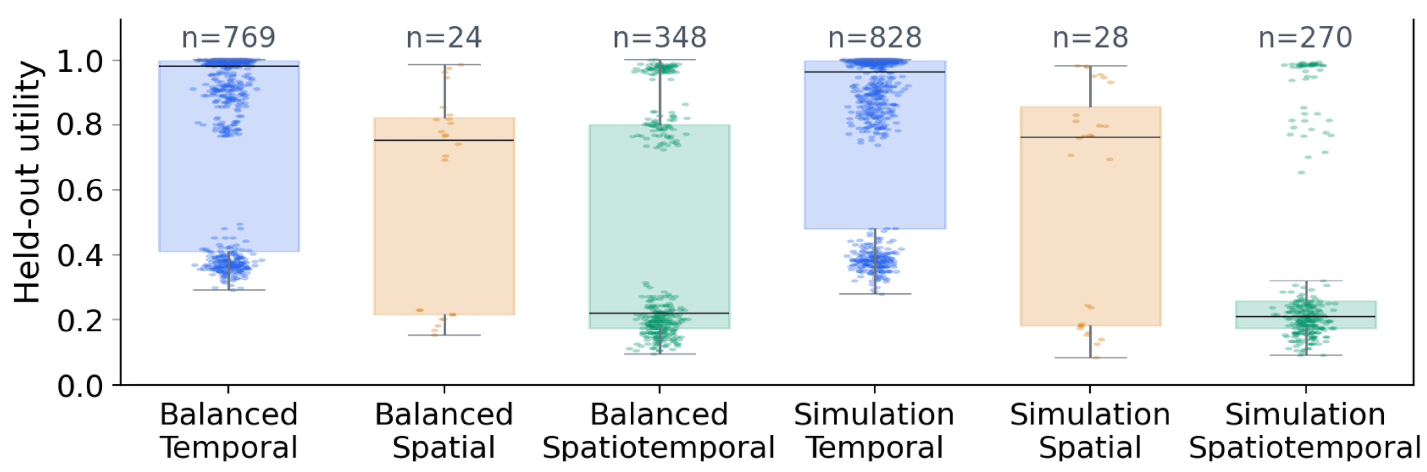
- **Evolutionary Loop:** Offline evolutionary loop on presimulated frozen history $\mathcal{H} = \mathcal{A}_{\text{train}} \cup \mathcal{A}_{\text{val}} \cup \mathcal{A}_{\text{test}}$

Aggregation-Only Study

- **Hypothesis H1: Decoupling from live simulation reduces temporal bias and increases spatial/mixed support.**
- **Experimental Programme:** Implemented experiments for aggregation-only study, transferable to other support modes
  - **Core Scenarios:** `Blinker` (positive control), `Mixed` (oscillators), `Noise` (negative control), `Soup` (random)
  - **CA Panel:** `Life` (B36/S23), `Diamoeba` (B35678/S5678), `Morley` (B36/S245), etc.
  - **Evaluation Profiles:** Replay (fidelity reference), Balanced Offline, Simulation-Oriented Offline
  - **Validation Readouts:** Same-seed reconstruction, null controls (time shuffle, space permute, phase randomize), cross-seed transfer, skip-ahead



Findings



- **Null control:** Offline evolution rejects structured-looking noise
- **Transferability:** Cross-seed transfer rewards simulation-oriented offline evolution
- **Hypothesis H1:** Decoupling reduces pressure towards temporal configurations — **but overrewarding skip-ahead reintroduces it.**

Contributions

- **Agnostic Observer Refinement:** Support formalism
- **Post-Hoc Protocol:** Frozen histories split into local anchors
- **Aggregation-Only Validation:** Successful rejection of noise and live-evaluation artifacts across four structural readouts
- **Generalizability:** Domain-agnostic framework extensible to non-CA systems & CA-based experimental programme

Future Work

- **Latent-Support Evaluation:** Studies including ORDER and GRAPH
- **Non-CA Domains:** Port formalism to continuous ABMs
- **Aperiodic Computation:** Extend periodic objective to capture gliders, attractors, and more

Acknowledgements

This research was partially funded by the Vogel-Stiftung Dr. Eckernkamp.

References

[1] James P. Crutchfield. The calculi of emergence: Computation, dynamics, and induction. *Physica D: Nonlinear Phenomena*, 75(1–3):11–54, 1994.
[2] Sebastian von Mammen. Evolving observers for agnostic pattern discovery in spatiotemporal fields. In *Proceedings of the 39th GI/ITG International Conference on Architecture of Computing Systems (ARCS)*, pages 84–100, Mainz, Germany, 2026. Springer.
[3] Claudio Angione, Eric Silverman, and Elisabeth Yaneske. Using machine learning as a surrogate model for agent-based simulations. *PLOS ONE*, 17(2):e0263150, 2022.